

MATHEMATICS

Chapter 1: Real Numbers



Real Numbers

Euclid's Division Lemma:

Given positive integers a and b , there exists unique integers q and r satisfying $a = bq + r$, where $0 \leq r < b$

➤ **Lemma** is a proven statement used for proving another statement.

Euclid's Division Lemma states that given two integers a and b , there exists a unique pair of integers q and r such that $a = b \times q + r$ and $0 \leq r < b$.

This lemma is essentially equivalent to: dividend = divisor $\times$ quotient + remainder

In other words, for a given pair of dividend and divisor, the quotient and remainder obtained are going to be unique.

Euclid's Division Algorithm:

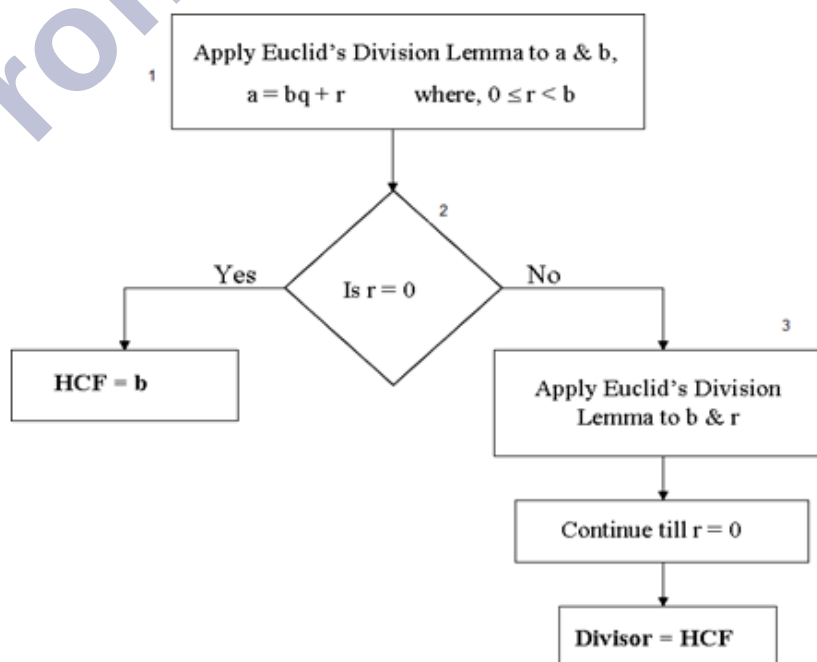
- An **algorithm** is a series of well defined steps which gives a procedure for solving a type of problem.
- This algorithm is a technique to compute the **H.C.F** of two given positive integers.
- According to this algorithm, the **HCF** of any two positive integers ' a ' and ' b ', with $a > b$, is obtained by following the steps given below:

Step 1: Apply Euclid's division lemma, to ' a ' and ' b ', to find q and r , such that $a = bq + r$, $0 \leq r < b$.

Step 2: If $r = 0$, the HCF is b . If $r \neq 0$, apply Euclid's division lemma to b and r .

Step 3: Continue the process till the remainder is zero. The divisor at this stage will be HCF (a, b). Also, note that $\text{HCF}(a, b) = \text{HCF}(b, r)$.

Euclid's Division Algorithm can be summarized as follows:



- Euclid's Division Algorithm is stated for only positive integers but it can be extended for all integers except zero, i.e., $b \neq 0$.

Consider two numbers 78 and 980 and we need to find the HCF of these numbers. To do this, we choose the largest integer first, i.e. 980 and then according to Euclid Division Lemma, $a = bq + r$ where $0 \leq r < b$;

$$980 = 78 \times 12 + 44$$

Now, here $a = 980$, $b = 78$, $q = 12$ and $r = 44$.

Now consider the divisor 78 and the remainder 44, apply Euclid division lemma again.

$$78 = 44 \times 1 + 34$$

Similarly, consider the divisor 44 and the remainder 34, apply Euclid division lemma to 44 and 34.

$$44 = 34 \times 1 + 10$$

Following the same procedure again,

$$34 = 10 \times 3 + 4$$

$$10 = 4 \times 2 + 2$$

$$4 = 2 \times 2 + 0$$

As we see that the remainder has become zero, therefore, proceeding further is not possible. Hence, the HCF is the divisor b left in the last step. We can conclude that the HCF of 980 and 78 is 2.

Let us try another example to find the HCF of two numbers 250 and 75. Here, the larger the integer is 250, therefore, by applying Euclid Division Lemma $a = bq + r$ where $0 \leq r < b$, we have

$$a = 250 \text{ and } b = 75$$

$$\Rightarrow 250 = 75 \times 3 + 25$$

By applying the Euclid's Division Algorithm to 75 and 25, we have:

$$75 = 25 \times 3 + 0$$

As the remainder becomes zero, we cannot proceed further. According to the algorithm, in this case, the divisor is 25. Hence, the HCF of 250 and 75 is 25.

Real Numbers:

- The numbers which can be represented in the form $\frac{p}{q}$, of where p and q are integers and $q \neq 0$ are called **Rational numbers**.
- Any number that cannot be expressed in the $\frac{p}{q}$, form of , where p and q are integers and $q \neq 0$ are called **Irrational numbers**.
- There are more irrational numbers than rational numbers between two consecutive numbers.

- Rational and Irrational numbers together constitute **Real numbers**.

Properties of Irrational numbers:

- The **Sum, Difference, Product** and **Division** of two irrational numbers need not always be an irrational number.
- Negative** of an irrational number is an irrational number.
- Sum** of a **rational** and an **irrational** number is irrational.
- Product** and **Division** of a non-zero rational and irrational number is always irrational.

Fractions:

- **Terminating fractions** are the fractions which leaves remainder 0 on normal division.
- **Recurring fractions** are the fractions which never leave a remainder 0 on normal division.

Properties related to prime numbers:

- If p is a prime and divides a^2 , then p divides a , where ' a ' is a positive integer.
- If p is a prime, then $\sqrt{p}$ is an irrational number.
- A number ends with the digit zero if and only if it has 2 and 5 as two of its prime factors.

1. Decimal Expansion:

- The decimal expansion of rational number is either **terminating** or **non-terminating recurring (repeating)**.
- If the decimal expansion of rational number **terminates**, then we can express the number in the form of $\frac{p}{q}$, where p and q are co-prime, and the prime factorization of q is of the form $2^n 5^m$, where n and m are non negative integers.
- If $x = \frac{p}{q}$ is a rational number, such that the prime factorization of q is of the form $2^n 5^m$, where n, m are non-negative integers. Then, x has a decimal expansion which **terminates**.
- If the denominator of a rational number is of the form $2^n 5^m$, then it will terminate after n places if $n > m$ or after m places if $m > n$.
- The decimal expansion of an irrational number is **non-terminating, non-recurring**.

Fundamental Theorem of Arithmetic:

Every composite number can be expressed (factorized) as a product of primes, and this factorization is unique, apart from the order in which the prime factors occur.

The procedure of finding **HCF (Highest Common Factor)** and **LCM (Lowest Common Multiple)** of given two positive integers a and b :

- i. Find the prime factorization of given numbers.
- ii. $\text{HCF}(a, b) = \text{Product of the smallest power of each common prime factors in the numbers.}$
- iii. $\text{LCM}(a, b) = \text{Product of the greatest power of each prime factors, involved in the numbers.}$

Fundamental Theorem of Arithmetic states that every integer greater than 1 is either a prime number or can be expressed in the form of primes. In other words, all the natural numbers can be expressed in the form of the product of its prime factors. To recall, prime factors are the numbers which are divisible by 1 and itself only. For example, the number 35 can be written in the form of its prime factors as:

$$35 = 7 \times 5$$

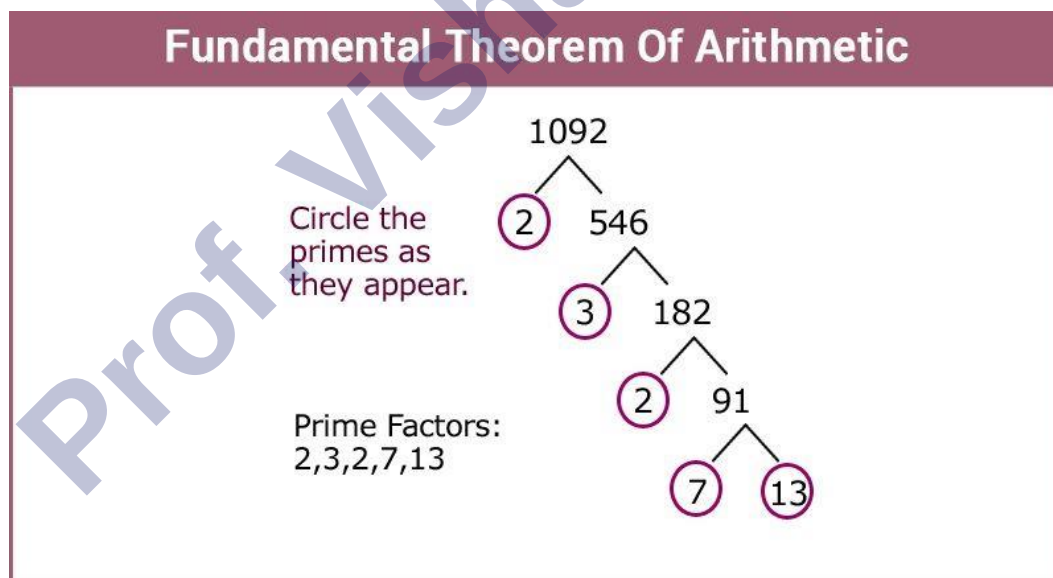
Here, 7 and 5 are the prime factors of 35

Similarly, another number 114560 can be represented as the product of its prime factors by using prime factorization method,

$$114560 = 2^7 \times 5 \times 179$$

So, we have factorized 114560 as the product of the power of its primes.

Therefore, every natural number can be expressed in the form of the product of the power of its primes. This statement is known as the Fundamental Theorem of Arithmetic, unique factorization theorem or the unique-prime-factorization theorem.



Proof for Fundamental Theorem of Arithmetic: In number theory, a composite number is expressed in the form of the product of primes and this factorization is unique apart from the order in which the prime factor occurs.

From this theorem we can also see that not only a composite number can be factorized as the product of their primes but also for each composite number the factorization is unique, not taking into consideration order of occurrence of the prime factors.

In simple words, there exists only a single way to represent a natural number by the

product of prime factors. This fact can also be stated as:

The prime factorization of any natural number is said to be unique for except the order of their factors.

In general, a composite number “a” can be expressed as,

$a = p_1 p_2 p_3 \dots p_n$, where $p_1, p_2, p_3 \dots p_n$ are the prime factors of a written in ascending order i.e. $p_1 \leq p_2 \leq p_3 \dots \leq p_n$.

Writing the primes in ascending order makes the factorization unique in nature.

Relationship between HCF and LCM of two numbers:

If a and b are two positive integers, then $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

Relationship between HCF and LCM of three numbers:

$$\text{LCM}(p, q, r) = \frac{p \cdot q \cdot r \cdot \text{HCF}(p, q, r)}{\text{HCF}(p, q) \cdot \text{HCF}(q, r) \cdot \text{HCF}(p, r)}$$

$$\text{HCF}(p, q, r) = \frac{p \cdot q \cdot r \cdot \text{LCM}(p, q, r)}{\text{LCM}(p, q) \cdot \text{LCM}(q, r) \cdot \text{LCM}(p, r)}$$

Method of Finding LCM

In Mathematics, the LCM of any two is the value that is evenly divisible by the two given numbers. The full form of LCM is Least Common Multiple. It is also called the Least Common Divisor (LCD). For example, $\text{LCM}(4, 5) = 20$. Here, the LCM 20 is divisible by both 4 and 5 such that 4 and 5 are called the divisors of 20.

LCM is also used to add or subtract any two fractions when the denominators of the fractions are different. While performing any arithmetic operations such as addition, subtraction with fractions, LCM is used to make the denominators common. This process makes the simplification process easier.

Least Common Multiple (LCM) is a method to find the smallest common multiple between any two or more numbers. A common multiple is a number which is a multiple of two or more numbers.

Properties of LCM

Properties	Description
Associative property	$\text{LCM}(a, b) = \text{LCM}(b, a)$
Commutative property	$\text{LCM}(a, b, c) = \text{LCM}(\text{LCM}(a, b), c) = \text{LCM}(a, \text{LCM}(b, c))$

Distributive property

$$\text{LCM}(da, db, dc) = d \text{ LCM}(a, b, c)$$

LCM Formula

Let a and b are two given integers. The formula to find the LCM of a & b is given by:

$$\text{LCM}(a, b) = (a \times b) / \text{GCD}(a, b)$$

Where $\text{GCD}(a, b)$ means Greatest Common Divisor or Highest Common Factor of a & b .

LCM Formula for Fractions

The formula to find the LCM of fractions is given by:

$$\text{L.C.M.} = \text{L.C.M Of Numerator} / \text{H.C.F Of Denominator}$$

Different Methods of LCM

There are three important methods by which we can find the LCM of two or more numbers. They are:

Listing the Multiples

Prime Factorization Method

Division Method

Listing the Multiples: The method to find the least common multiple of any given numbers is first to list down the multiples of specific numbers and then find the first common multiple between them.

Suppose there are two numbers 11 and 33. Then by listing the multiples of 11 and 33, we get;

Multiples of 11 = 11, 22, 33, 44, 55,

Multiples of 33 = 33, 66, 99,

We can see, the first common multiple or the least common multiple of both the numbers is 33. Hence, the $\text{LCM}(11, 33) = 33$.

LCM By Prime Factorization: Another method to find the LCM of the given numbers is prime factorization. Suppose there are three numbers 12, 16 and 24. Let us write the prime factors of all three numbers individually.

$$12 = 2 \times 2 \times 3$$

$$16 = 2 \times 2 \times 2 \times 2$$

$$24 = 2 \times 2 \times 2 \times 3$$

Now writing the prime factors of all the three numbers together, we get;

$$12 \times 16 \times 24 = 2 \times 2 \times 3 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3$$

Now pairing the common prime factors we get the LCM. Hence, there are four 2's and one 3. So the LCM of 12, 16 and 24 will be;

$$\text{LCM}(12, 16, 24) = 2 \times 2 \times 2 \times 2 \times 3 = 48$$

LCM By Division Method

- Finding LCM of two numbers by division method is an easy method. Below are the steps to find the LCM by division method:
- First, write the numbers, separated by commas
- Now divide the numbers, by the smallest prime number.
- If any number is not divisible, then write down that number and proceed further
- Keep on dividing the row of numbers by prime numbers, unless we get the results as 1 in the complete row
- Now LCM of the numbers will be equal to the product of all the prime numbers we obtained in the division method

Example: To find the Least Common Multiple (L.C.M) of 36 and 56,

$$36 = 2 \times 2 \times 3 \times 3$$

$$56 = 2 \times 2 \times 2 \times 7$$

The common prime factors are 2×2

The uncommon prime factors are 3×3 for 36 and 2×7 for 56.

LCM of 36 and 56 = $2 \times 2 \times 3 \times 3 \times 2 \times 7$ which is 504

Method of Finding HCF

H.C.F can be found using two methods – Prime factorisation and Euclid's division algorithm.

Prime Factorisation: Given two numbers, we express both of them as products of their respective prime factors. Then, we select the prime factors that are common to both the numbers

Example – To find the H.C.F of 20 and 24

$$20 = 2 \times 2 \times 5 \text{ and } 24 = 2 \times 2 \times 2 \times 3$$

The factor common to 20 and 24 is 2×2 , which is 4, which in turn is the H.C.F of 20 and 24.

Euclid's Division Algorithm: It is the repeated use of Euclid's division lemma to find the H.C.F of two numbers.

Example: To find the HCF of 18 and 30

Steps to find the HCF of any given numbers.

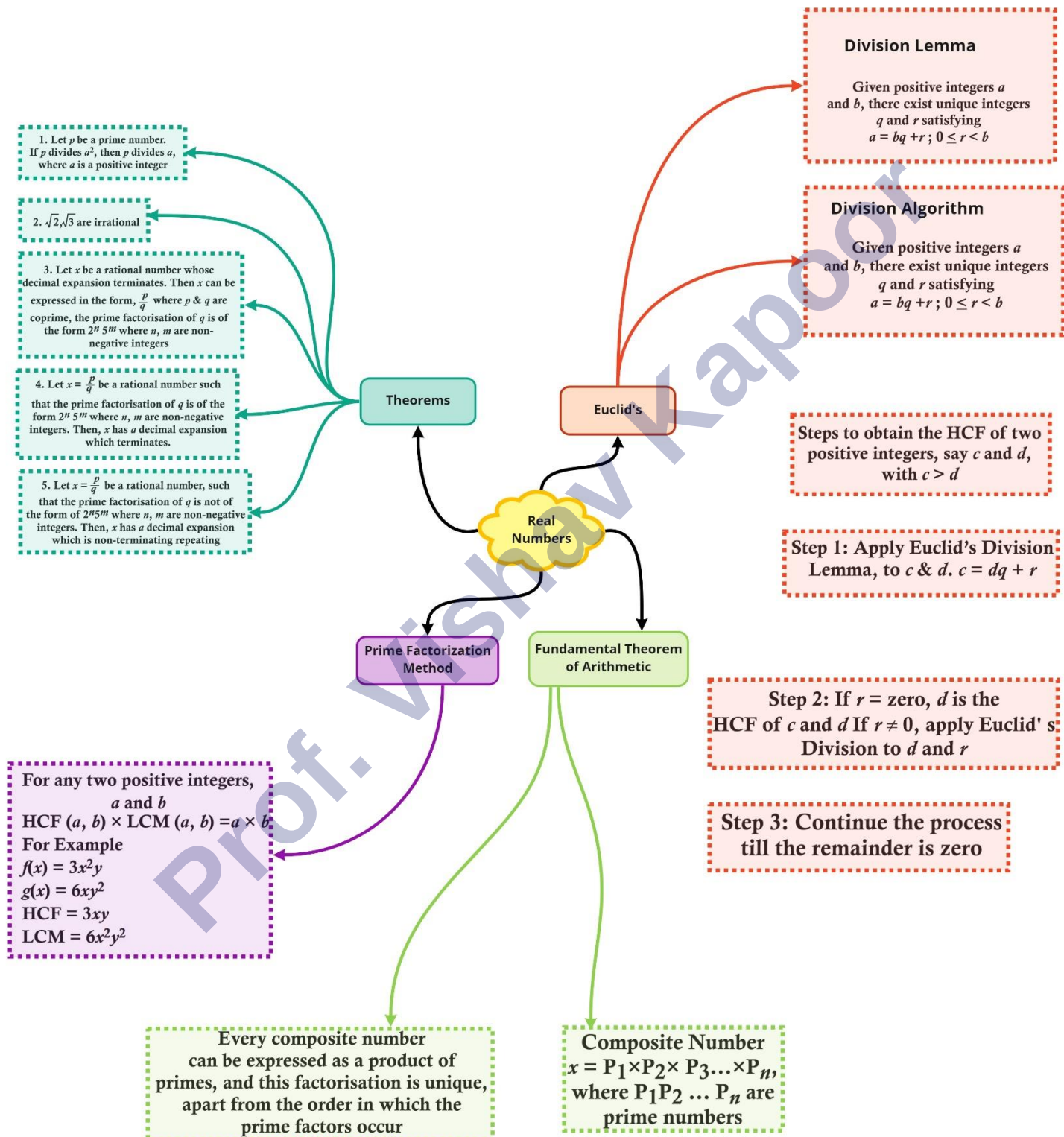
Larger Number/Smaller Number

Divisor of step 1/Remainder

Divisor of step 2/Remainder

Step 5: The divisor of the last step is the HCF.

Class : 10th mathematics
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Important Questions

Multiple Choice questions-

1. HCF of 8, 9, 25 is

- (a) 8
- (b) 9
- (c) 25
- (d) 1

2. Which of the following is not irrational?

- (a) $(2 - \sqrt{3})^2$
- (b) $(\sqrt{2} + \sqrt{3})^2$
- (c) $(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})$
- (d) $\frac{2\sqrt{7}}{7}$

3. The product of a rational and irrational number is

- (a) rational
- (b) irrational
- (c) both of above
- (d) none of above

4. The sum of a rational and irrational number is

- (a) rational
- (b) irrational
- (c) both of above
- (d) none of above

5. The product of two different irrational numbers is always

- (a) rational

(b) irrational

(c) both of above

(d) none of above

6. The sum of two irrational numbers is always

(a) irrational

(b) rational

(c) rational or irrational

(d) one

7. If $b = 3$, then any integer can be expressed as $a =$

(a) $3q, 3q + 1, 3q + 2$

(b) $3q$

(c) none of the above

(d) $3q + 1$

8. The product of three consecutive positive integers is divisible by

(a) 4

(b) 6

(c) no common factor

(d) only 1

9. The set $A = \{0, 1, 2, 3, 4, \dots\}$ represents the set of

(a) whole numbers

(b) integers

(c) natural numbers

(d) even numbers

10. Which number is divisible by 11?

(a) 1516

(b) 1452

(c) 1011

(d) 1121

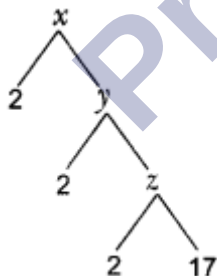
Very Short Questions:

1. What is the HCF of the smallest composite number and the smallest prime number?
2. The decimal representation of $\frac{6}{1250}$ will terminate after how many places of decimal?
3. If HCF of a and b is 12 and product of these numbers is 1800. Then what is LCM of these numbers?
4. What is the HCF of $3^3 \times 5$ and $3^2 \times 5^2$?
5. If a is an odd number, b is not divisible by 3 and LCM of a and b is P, what is the LCM of 3a and 2b?
6. If P is prime number then, what is the LCM of P, P^2 , P^3 ?
7. Two positive integers p and q can be expressed as $p = ab^2$ and $q = a^2b$, a and b are prime numbers. What is the LCM of p and q?
8. A number N when divided by 14 gives the remainder 5. What is the remainder when the same number is divided by 7?
9. Examine whether $\frac{17}{30}$ is a terminating decimal or not.
10. What are the possible values of remainder r, when a positive integer a is divided by 3?
11. A rational number in its decimal expansion is 1.7351. What can you say about the prime factors of q when this number is expressed in the form $\frac{p}{q}$? Give reason.
12. Without actually performing the long division, find $\frac{987}{10500}$ will have terminating or non-terminating repeating decimal expansion. Give reason for your answer.

Short Questions :

1. Can the number 4^n , n be a natural number, end with the digit 0? Give reason.
2. Write whether the square of any positive integer can be of the form $3m + 2$, where m is a natural number. Justify your answer.

3. Can two numbers have 18 as their HCF and 380 as their LCM? Give reason.
4. An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march?
5. Find the LCM and HCF of 12, 15 and 21 by applying the prime factorisation method.
6. Find the LCM and HCF of the following pairs of integers and verify that $\text{LCM} \times \text{HCF} = \text{product of the two numbers}$.
(1) 26 and 91 (ii) 198 and 144
7. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same. Suppose they both start from the same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?
8. Write down the decimal expansions of the following numbers:
(i) $\frac{35}{50}$ (ii) $\frac{15}{1600}$
9. Express the number $\overline{0.3178}$ in the form of rational number $\frac{a}{b}$.
10. If n is an odd positive integer, show that $(n^2 - 1)$ is divisible by 8.
11. The LCM of two numbers is 14 times their HCF. The sum of LCM and HCF is 600. If one number is 280, then find the other number.
12. Find the value of x , y and z in the given factor tree. Can the value of ' x ' be found without finding the value of ' y ' and ' z '? If yes, explain.



13. Show that any positive odd integer is of the form $6q + 1$ or $6q + 3$ or $6q + 5$ where q is some integer.
14. The decimal expansions of some real numbers are given below. In each case, decide whether they are rational or not. If they are rational, write it in the form $\frac{p}{q}$. What can you say about the prime factors of q ?

(i) $0.140140014000140000\dots$ (ii) $\overline{0.16}$

Long Questions :

1. Use Euclid's division lemma to show that the square of any positive integer is either of the form $3m$ or $3m + 1$ for some integer m .
2. Show that one and only one out of n , $n + 2$, $n + 4$ is divisible by 3, where n is any positive integer.
3. Use Euclid's division algorithm to find the HCF of:
(i) 960 and 432
(ii) 4052 and 12576.
4. Using prime factorisation method, find the HCF and LCM of 30, 72 and 432. Also show that $\text{HCF} \times \text{LCM} \neq \text{Product of the three numbers}$.
5. Prove that $\sqrt{7}$ is an irrational number.
6. Show that $5 - \sqrt{3}$ is an irrational number.
7. Using Euclid's division algorithm, find whether the pair of numbers 847, 2160 are co-prime or not.
8. Check whether 6^n can end with the digit 0 for any natural number n .
9. Show that there is no positive integer n for which $\sqrt{n-1} + \sqrt{n+1}$ is rational.
10. Find the largest positive integer that will divide 398, 436 and 542 leaving remainders 7, 11 and 15 respectively.

Case Study Questions:

1. Srikanth has made a project on real numbers, where he finely explained the applicability of exponential laws and divisibility conditions on real numbers. He also included some assessment questions at the end of his project as listed below. Answer them.
 - i. For what value of n , $4n$ ends in 0?
 - a. 10
 - b. When n is even.
 - c. When n is odd.
 - d. No value of n .

ii. If a is a positive rational number and n is a positive integer greater than 1, then for what value of n , $4n$ is a rational number?

- a. When n is any even integer.
- b. When n is any odd integer.
- c. For all $n > 1$.
- d. Only when $n = 0$.

iii. If x and y are two odd positive integers, then which of the following is true?

- a. $x^2 + y^2$ is even.
- b. $x^2 + y^2$ is not divisible by 4.
- c. $x^2 + y^2$ is odd.
- d. Both (a) and (b).

iv. The statement 'One of every three consecutive positive integers is divisible by 3' is:

- a. Always true.
- b. Always false.
- c. Sometimes true.
- d. None of these.

v. If n is any odd integer, then $n^2 - 1$ is divisible by:

- a. 22
- b. 55
- c. 88
- d. 8

2. Real numbers are extremely useful in everyday life. That is probably one of the main reasons we all learn how to count and add and subtract from a very young age. Real numbers help us to count and to measure out quantities of different items in various fields like retail, buying, catering, publishing etc. Every normal person uses real numbers in his daily life. After knowing the importance of real numbers, try and improve your knowledge about them by answering the following questions on real life based situations.

- i. Three people go for a morning walk together from the same place. Their steps measure 80cm, 85cm and 90cm respectively. What is the minimum distance travelled when they meet at first time after starting the walk assuming that their walking speed is same?
- 6120cm
 - 12240cm
 - 4080cm
 - None of these
- ii. In a school Independence Day parade, a group of 594 students need to march behind a band of 189 members. The two groups have to march in the same number of columns. What is the maximum number of columns in which they can march?
- 9
 - 6
 - 27
 - 29
- iii. Two tankers contain 768 litres and 420 litres of fuel respectively. Find the maximum capacity of the container which can measure the fuel of either tanker exactly.
- 4 litres
 - 7 litres
 - 12 litres
 - 18 litres
- iv. The dimensions of a room are 8m, 25cm, 6m, 75cm and 4m, 50cm. Find the length of the largest measuring rod which can measure the dimensions of room exactly.
- 1m, 25cm
 - 75cm
 - 90cm
 - 1m, 35cm
- v. Pens are sold in pack of 8 and notepads are sold in pack of 12. Find the least number of pack of each type that one should buy so that there are equal number of pens and notepads.
- 3 and 2
 - 2 and 5
 - 3 and 4

d. 4 and 5

Assertion Reason Questions-

1. **Directions:** In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true and R is not the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

Assertion: $11 \times 4 \times 3 \times 2 + 4$ is a composite number.

Reason: Every composite number can be expressed as product of primes.

2. **Directions:** In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true and R is not the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

Assertion: If $\text{LCM} = 350$, product of two numbers is 25×70 , then their $\text{HCF} = 5$

Reason: $\text{LCM} \times \text{product of numbers} = \text{HCF}$

Answer Key-**Multiple Choice questions-**

1. (d) 1
2. (c) $(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})$
3. (b) irrational
4. (b) irrational
5. (b) irrational
6. (a) irrational
7. (a) $3q, 3q+1, 3q+2$
8. (b) 6
9. (a) whole numbers
10. (b) 1452

Very Short Answer :

1. Smallest composite number = 4

Smallest prime number = 2

So, HCF (4, 2) = 2

- 2.

$$\frac{6}{1250} = \frac{3}{625} = \frac{3}{5^4} \times \frac{2^4}{2^4} = \frac{48}{(5 \times 2)^4} = \frac{48}{10^4} = 0.0048$$

This representation will terminate after 4 decimal places.

3. Product of two numbers = Product of their LCM and HCF

$$\Rightarrow 1800 = 12 \times \text{LCM}$$

$$\Rightarrow \text{LCM} = \frac{1800}{12} = 150.$$

4. HCF of $3^3 \times 5$ and $3^2 \times 5^2 = 3^2 \times 5 = 45$

5. 6P

6. p^3

7. a^2h^2

8. 5, because 14 is multiple of 7.

Therefore, remainder in both cases are same.

9.

$$\frac{17}{30} = \frac{17}{2 \times 3 \times 5}$$

Since the denominator has 3 as its factor.

$\therefore \frac{17}{30}$ is a non-terminating decimal.

10. According to Euclid's division lemma

$a = 3q + r$, where $0 \leq r < 3$ and r is an integer.

Therefore, the values of r can be 0, 1 or 2.

11. As 1.7351 is a terminating decimal number, so q must be of the form $2^m 5^n$, where m, n are natural numbers.

12. $\frac{987}{10500} = \frac{47}{500}$ and $500 = 2^2 \times 5^3$, so it has terminating decimal expansion.

Short Answer :

1. If 4^n ends with 0, then it must have 5 as a factor. But, $(4)^n = (2^2)^n = 2^{2n}$ i.e., the only prime factor

of 4^n is 2. Also, we know from the fundamental theorem of arithmetic that the prime factorization of each number is unique.

$\therefore 4^n$ can never end with 0.

2. No, because any positive integer can be written as $3q, 3q + 1, 3q + 2$, therefore, square will be

$$9q^2 = 3m, 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1 = 3m + 1,$$

$$9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1 = 3m + 1.$$

3. No, because here HCF (18) does not divide LCM (380).

4. For the maximum number of columns, we have to find the HCF of 616 and 32.

Now, since $616 > 32$, we apply division lemma to 616 and 32.

We have, $616 = 32 \times 19 + 8$

Here, remainder $8 \neq 0$. So, we again apply division lemma to 32 and 8.

We have, $32 = 8 \times 4 + 0$

Here, remainder is zero. So, $\text{HCF}(616, 32) = 8$

Hence, maximum number of columns is 8.

5. The prime factors of 12, 15 and 21 are

$$12 = 2^2 \times 3, 15 = 3 \times 5 \text{ and } 21 = 3 \times 7$$

Therefore, the HCF of these integers is 3.

$2^2, 3^1, 5^1$ and 7^1 are the greatest powers involved in the prime factors of 12, 15 and 21.

$$\text{So, LCM}(12, 15, 21) = 2^2 \times 3^1 \times 5^1 \times 7^1 = 420.$$

6. (i) We have, $26 = 2 \times 13$ and $91 = 7 \times 13$

$$\text{Thus, LCM}(26, 91) = 2 \times 7 \times 13 = 182$$

$$\text{HCF}(26, 91) = 13$$

$$\text{Now, LCM}(26, 91) \times \text{HCF}(26, 91) = 182 \times 13 = 2366$$

$$\text{and Product of the two numbers} = 26 \times 91 = 2366$$

Hence, $\text{LCM} \times \text{HCF} = \text{Product of two numbers}$.

$$(ii) 144 = 24 \times 32 \text{ and } 198 = 2 \times 32 \times 11$$

$$\therefore \text{LCM}(198, 144) = 24 \times 32 \times 11 = 1584$$

$$\text{HCF}(198, 144) = 2 \times 32 = 18$$

$$\text{Now, LCM}(198, 144) \times \text{HCF}(198, 144) = 1584 \times 18 = 28512$$

$$\text{and product of 198 and 144} = 28512$$

$$\text{Thus, product of LCM}(198, 144) \text{ and HCF}(198, 144)$$

$$= \text{Product of 198 and 144}.$$

7. To find the time after which they meet again at the starting point, we have to find LCM of 18 and 12 minutes. We have

$$\begin{array}{c|c} 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\begin{array}{c|c} 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

Therefore, LCM of 18 and 12 = $2^2 \times 3^2 = 36$

So, they will meet again at the starting point after 36 minutes.

8. (i)

$$\begin{aligned} \text{We have, } \frac{35}{50} &= \frac{35}{5^2 \times 2} = \frac{35 \times 2}{5^2 \times 2 \times 2} = \frac{70}{5^2 \times 2^2} \\ &= \frac{70}{10^2} = \frac{70}{100} = 0.70 \end{aligned}$$

(ii)

$$\begin{aligned} \text{We have, } \frac{15}{1600} &= \frac{15}{2^6 \times 5^2} = \frac{15 \times 5^4}{2^4 \times 2^2 \times 5^2 \times 5^4} = \frac{15 \times 625}{2^6 \times 5^6} \\ &= \frac{9375}{10^6} = \frac{9375}{1000000} = 0.009375 \end{aligned}$$

9. Let $x = \overline{0.3178}$

then $x = 0.3178178178 \dots \dots$ (i)

$10x = 3.178178178 \dots \dots$ (ii)

$10000x = 3178.178178 \dots \dots$ (iii)

On subtracting (ii) from (iii), we get

$$9990x = 3175 \Rightarrow x = \frac{3175}{9990} = \frac{635}{1998}$$

$$\therefore \overline{0.3178} = \frac{635}{1998}$$

10. We know that an odd positive integer n is of the form $(4q + 1)$ or $(4 + 3)$ for some integer q .

Case – I When $n = (4q + 1)$

$$\text{In this case } n^2 - 1 = (4q + 1)^2 - 1 = 16q^2 + 8q = 8q(2q + 1)$$

which is clearly divisible by 8.

Case – II When $n = (4q + 3)$

In this case, we have

$$n^2 = (4q + 3)^2 - 1 = 16q^2 + 24q + 8 = 8(2q^2 + 3q + 1)$$

which is clearly divisible by 8.

Hence $(n^2 - 1)$ is divisible by 8.

11. Let HCF of the numbers be x then according to question LCM of the number will be $14x$

$$\text{And } x + 14 = 600 \Rightarrow 15x = 600 \Rightarrow x = 40$$

$$\text{Then HCF} = 40 \text{ and LCM} = 14 \times 40 = 560$$

$$\therefore \text{LCM} \times \text{HCF} = \text{Product of the numbers}$$

$$560 \times 40 = 280 \times \text{Second number} \quad \text{Second number} = \frac{560 \times 40}{280} = 80$$

Then other number is 80.

12. $z = 2 \times 17 = 34$; $y = 34 \times 2 = 68$ and $x = 2 \times 68 = 136$

Yes, value of x can be found without finding value of y or z as

$x = 2 \times 2 \times 2 \times 17$ which are prime factors of x .

13. Let a be any positive odd integer and $h = 6$. Then, by Euclid's algorithm, $a = 6q + r$, for some

integer $q \geq 0$ and $0 \leq r < 6$.

i.e., the possible remainders are 0, 1, 2, 3, 4, 5.

Thus, a can be of the form $6q$, or $6q + 1$, or $6q + 2$, or $6q + 3$, or $6q + 4$,

or $6q + 5$, where q is some quotient.

Since a is odd integer, so a cannot be of the form $6q$, or $6q + 2$, or $6q + 4$, (since they are even).

Thus, a is of the form $6q + 1$, $6q + 3$, or $6q + 5$, where q is some integer.

Hence, any odd positive integer is of the form $6q + 1$ or $6q + 3$ or $6q + 5$, where q is some integer.

14. (i) We have, 0.140140014000140000... a non-terminating and non-repeating

decimal expansion. So it is irrational. It cannot be written in the form of $\frac{p}{q}$

(ii) We have, $\overline{0.16}$ a non-terminating but repeating decimal expansion. So it is rational.

$$\text{Let } x = \overline{0.16}$$

$$\text{Then, } x = 0.1616... \text{ (i)}$$

$$100x = 16.1616... \text{ (ii)}$$

On subtracting (i) from (ii), we get

$$100x - x = 16.1616 - 0.1616$$

$$\Rightarrow 99x = 16 \Rightarrow x = \frac{16}{99} = \frac{p}{q}$$

The denominator (q) has factors other than 2 or 5.

Long Answer :

- Let a be an arbitrary positive integer.

Then by Euclid's division algorithm, corresponding to the positive integers a and 3 there exist

non-negative integers q and r such that

$$a = 3q + r \text{ where } 0 \leq r < 3$$

$$a^2 = 9q^2 + 6qr + r^2 \dots (i) \quad 0 \leq r < 3$$

Case – I: When $r = 0$ [putting in (i)]

$$a^2 = 9q^2 = 3(3q^2) = 3m \text{ where } m = 3q^2$$

Case – II: $r = 1$

$$a^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1 = 3m + 1 \text{ where } m = 3q^2 + 2q$$

Case – III: $r = 2$

$$a^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1 = 3m + 1 \text{ where } m = (3q^2 + 4q + 1)$$

Hence, square of any positive integer is either of the form $3m$ or $3m + 1$ for some integer m.

- Let q be the quotient and r be the remainder when n is divided by 3.

Therefore, $n = 3q + r$, where $r = 0, 1, 2$

$n = 3q$ or $n = 3q + 1$ or $n = 3q + 2$

Case (i) if $n = 3q$, then n is divisible by 3, $n + 2$ and $n + 4$ are not divisible by 3.

Case (ii) if $n = 3q + 1$ then $n + 2 = 3q + 3 = 3(q + 1)$, which is divisible by 3 and

$n + 4 = 3q + 5$, which is not divisible by 3.

So, only $(n + 2)$ is divisible by 3.

Case (iii) If $n = 3q + 2$, then $n + 2 = 3q + 4$, which is not divisible by 3 and

$(n + 4) = 3q + 6 = 3(q + 2)$, which is divisible by 3.

So, only $(n + 4)$ is divisible by 3.

Hence one and only one out of n , $(n + 2)$, $(n + 4)$, is divisible by 3.

3. (j) Since $960 > 432$, we apply the division lemma to 960 and 432.

We have, $960 = 432 \times 2 + 96$

Since the remainder $96 \neq 0$, so we apply the division lemma to 432 and 96.

We have, $432 = 96 \times 4 + 48$

Again remainder $48 \neq 0$ so we again apply division lemma to 96 and 48.

We have, $96 = 48 \times 2 + 0$

The remainder has now become zero. So our procedure stops.

Since the divisor at this stage is 48.

Hence, HCF of 960 and 432 is 48.

i.e., $\text{HCF}(960, 432) = 48$

- (ii) Since $12576 > 4052$, we apply the division lemma to 12576 and 4052, to get

$12576 = 4052 \times 3 + 420$

Since the remainder $420 \neq 0$, we apply the division lemma to 4052 and 420, to get

$4052 = 420 \times 9 + 272$

We consider the new divisor 420 and the new remainder 272, and apply the division lemma to get

$$420 = 272 \times 1 + 148$$

We consider the new divisor 272 and the new remainder 148, and apply the division lemma to get

$$272 = 148 \times 1 + 124$$

We consider the new divisor 148 and the new remainder 124, and apply the division lemma to get

$$148 = 124 \times 1 + 24$$

We consider the new divisor 124 and the new remainder 24, and apply the division lemma to get

$$124 = 24 \times 5 + 4$$

We consider the new divisor 24 and the new remainder 4, and apply the division lemma to get

$$24 = 4 \times 6 + 0$$

The remainder has now become zero, so our procedure stops. Since the divisor at this stage is 4, the HCF of 12576 and 4052 is 4.

4. Given members = 30, 72, 432 .

$$30 = 2 \times 3 \times 5; 72 = 2^3 \times 3^2 \text{ and } 432 = 2^4 \times 3^3$$

Here, 2^1 and 3^1 are the smallest powers of the common factors 2 and 3 respectively.

$$\text{So, HCF } (30, 72, 432) = 2^1 \times 3^1 = 2 \times 3 = 6$$

Again, 2^4 , 3^3 and 5^1 are the greatest powers of the prime factors 2, 3 and 5 respectively.

$$\text{So, LCM } (30, 72, 432) = 2^4 \times 3^3 \times 5^1 = 2160$$

$$\text{HCF} \times \text{LCM} = 6 \times 2160 = 12960$$

$$\text{Product of numbers} = 30 \times 72 \times 432 = 933120 .$$

Therefore, $\text{HCF} \times \text{LCM} \neq \text{Product of the numbers}.$

5. Let us assume, to the contrary, that $\sqrt{7}$ is a rational number.

Then, there exist co-prime positive integers a and b such that

$$\sqrt{7} = \frac{a}{b}, b \neq 0$$

So, $a = \sqrt{7} b$

Squaring both sides, we have

$$a^2 = 7b^2 \dots\dots (i)$$

$$\Rightarrow 7 \text{ divides } a^2 \Rightarrow 7 \text{ divides } a$$

So, we can write

$$a = 7c \text{ (where } c \text{ is an integer)}$$

Putting the value of $a = 7c$ in (i), we have

$$49c^2 = 7b^2 \quad 7^2 = b^2$$

It means 7 divides b^2 and so 7 divides b .

So, 7 is a common factor of both a and b which is a contradiction.

So, our assumption that $\sqrt{7}$ is rational is wrong.

Hence, we conclude that $\sqrt{7}$ is an irrational number.

6. Let us assume that $5 - \sqrt{3}$ is rational.

So, $5 - \sqrt{3}$ may be written as

$$5 - \sqrt{3} = \frac{p}{q}, \text{ where } p \text{ and } q \text{ are integers, having no common factor except 1 and } q \neq 0.$$

$$\Rightarrow 5 - \frac{p}{q} = \sqrt{3} \Rightarrow \sqrt{3} = \frac{5q-p}{q}$$

Since $\frac{5q-p}{q}$ is a rational number as p and q are integers.

$\therefore \sqrt{3}$ is also a rational number which is a contradiction.

Thus, our assumption is wrong.

Hence, $5 - \sqrt{3}$ is an irrational number.

7. Since $2160 > 847$ we apply the division lemma to 2160 and 847

$$\text{we have, } 2160 = 847 \times 2 + 466$$

Since remainder $466 \neq 0$. So, we apply the division lemma to 847 and 466

$$847 = 466 \times 1 + 381$$

Again remainder $381 \neq 0$. So we again apply the division lemma to 466 and 381.

$$466 = 381 \times 1 + 85$$

Again remainder $85 \neq 0$. So, we again apply the division lemma to 381 and 85

$$381 = 85 \times 4 + 41$$

Again remainder $41 \neq 0$. So, we again apply the division lemma to 85 and 41.

$$85 = 41 \times 2 + 3$$

Again remainder $3 \neq 0$. So, we again apply the division lemma to 41 and 3.

$$41 = 3 \times 13 + 2$$

Again remainder $2 \neq 0$. So, we again apply the division lemma to 3 and 2.

$$3 = 2 \times 1 + 1$$

Again remainder $1 \neq 0$. So, we apply division lemma to 2 and 1

$$2 = 1 \times 2 + 0$$

The remainder now becomes 0. So, our procedure stops.

Since the divisor at this stage is 1.

Hence, HCF of 847 and 2160 is 1 and numbers are co-prime.

8. If the number 6^n , for any n , were to end with the digit zero, then n would be divisible by 5. That is, the prime factorisation of 6^n would contain the prime 5. But $6^n = (2 \times 3)^n = 2^n \times 3^n$. So the primes in factorisation of 6^n are 2 and 3. So the uniqueness of the Fundamental Theorem of Arithmetic guarantees that (here are no other primes except 2 and 3 in the factorisation of 6^n . So there is no natural number n for which 6^n ends with digit zero.
9. Let there be a positive integer n for which $\sqrt{n-1} + \sqrt{n+1}$ be rational number.

$$\sqrt{n-1} + \sqrt{n+1} = \frac{p}{q}; \text{ where } p, q \text{ are integers and } q \neq 0 \quad \dots(i)$$

$$\Rightarrow \frac{1}{\sqrt{n-1} + \sqrt{n+1}} = \frac{q}{p} \quad \Rightarrow \frac{\sqrt{n-1} - \sqrt{n+1}}{(\sqrt{n-1} + \sqrt{n+1}) \times (\sqrt{n-1} - \sqrt{n+1})} = \frac{q}{p}$$

$$\Rightarrow \frac{\sqrt{n-1} - \sqrt{n+1}}{(n-1) - (n+1)} = \frac{q}{p} \quad \Rightarrow \frac{\sqrt{n-1} - \sqrt{n+1}}{n-1-n-1} = \frac{q}{p}$$

$$\Rightarrow \frac{\sqrt{n+1} - \sqrt{n-1}}{2} = \frac{q}{p} \quad \Rightarrow \sqrt{n+1} - \sqrt{n-1} = \frac{2q}{p} \quad \dots(ii)$$

Adding (i) and (ii), we get

$$\begin{aligned} \sqrt{n-1} + \sqrt{n+1} + \sqrt{n+1} - \sqrt{n-1} &= \frac{p}{q} + \frac{2q}{p} \\ \Rightarrow 2\sqrt{n+1} &= \frac{p^2 + 2q^2}{pq} \quad \Rightarrow \sqrt{n+1} = \frac{p^2 + 2q^2}{2pq} \\ \Rightarrow \sqrt{n+1} \text{ is rational number as } \frac{p^2 + 2q^2}{2pq} \text{ is rational} \\ \Rightarrow \sqrt{n+1} \text{ is perfect square of positive integer} \quad \dots(A) \end{aligned}$$

Again subtracting (ii) from (i), we get

$$\begin{aligned} \sqrt{n-1} + \sqrt{n+1} - \sqrt{n+1} + \sqrt{n-1} &= \frac{p}{q} - \frac{2q}{p} \Rightarrow 2\sqrt{n-1} = \frac{p^2 - 2q^2}{pq} \\ \Rightarrow \sqrt{n-1} \text{ is rational number as } \frac{p^2 - 2q^2}{2pq} \text{ is rational.} \end{aligned}$$

$\Rightarrow \sqrt{n-1}$ is also perfect square of positive integer From (A) and (B)

$\sqrt{n+1}$ and $\sqrt{n-1}$ are perfect squares of positive integer. It contradicts the fact that two perfect squares differ at least by 3.

Hence, there is no positive integer n for which $\sqrt{n-1} + \sqrt{n+1}$ is rational.

10. It is given that on dividing 398 by the required number, there is a remainder of 7. This means that $398 - 7 = 391$ is exactly divisible by the required number. In other words, required number is a factor of 391.

Similarly, required positive integer is a factor of $436 - 11 = 425$ and $542 - 15 = 527$

Clearly, the required number is the HCF of 391, 425 and 527.

Using the factor tree, we get the prime factorisations of 391, 425 and 527 as follows:

$$391 = 17 \times 23, 425 = 5^2 \times 17 \text{ and } 527 = 17 \times 31$$

$\therefore$ HCF of 391, 425, and 527 is 17.

Hence, the required number = 17.

Case Study Answers:

1. Answer :

- i. (d) No value of n .

Solution:

For a number to end in zero it must be divisible by 5, but $4^n = 2^{2n}$ is never divisible by 5. So, 4^n never ends in zero for any value of n .

- ii. (c) For all $n > 1$.

Solution:

We know that product of two rational numbers is also a rational number.

So, $a^2 = a \times a = \text{rational number}$.

$a^3 = a^2 \times a = \text{rational number}$.

$a^4 = a^3 \times a = \text{rational number}$.

.....

.....

$a^n = a^{n-1} \times a = \text{rational number}$.

- iii. (d) Both (a) and (b).

Solution:

Let $x = 2m + 1$ and $y = 2k + 1$

Then $x^2 + y^2 = (2m + 1)^2 + (2k + 1)^2$

$= 4m^2 + 4m + 1 + 4k^2 + 4k + 1$

$= 4(m^2 + k^2 + m + k) + 2$

So, it is even but not divisible by 4.

- iv. (a) Always true.

Solution:

Let three consecutive positive integers be n , $n + 1$ and $n + 2$.

We know that when a number is divided by 3, the remainder obtained is either 0 or 1 or 2.

So, $n = 3p$ or $3p + 1$ or $3p + 2$, where p is some integer.

If $n = 3p$, then $n + 2$ is divisible by 3.

If $n = 3p + 1$, then $n + 2 = 3p + 1 + 2 = 3p + 3 = 3(p + 1)$ is divisible by 3.

If $n = 3p + 2$, then $n + 1 = 3p + 2 + 1 = 3p + 3 = 3(p + 1)$ is divisible by 3.

So, we can say that one of the numbers among n , $n + 1$ and $n + 2$ is always divisible by 3.

v. (d) 8

Solution:

Any odd number is of the form of $(2k + 1)$, where k is any integer.

So, $n^2 - 1 = (2k + 1)^2 - 1 = 4k^2 + 4k$

For $k = 1$, $4k^2 + 4k = 8$, which is divisible by 8.

Similarly, for $k = 2$, $4k^2 + 4k = 24$, which is divisible by 8.

And for $k = 3$, $4k^2 + 4k = 48$, which is also divisible by 8.

So, $4k^2 + 4k$ is divisible by 8 for all integers k , i.e., $n^2 - 1$ is divisible by 8 for all odd values of n .

2. Answer :

i. (b) 12240cm

Solution:

Here $80 = 24 \times 5$, $85 = 17 \times 5$

and $90 = 2 \times 32 \times 5$

L.C.M of 80, 85 and 90 = $24 \times 3 \times 3 \times 5 \times 17 = 12240$

Hence, the minimum distance each should walk when they at first time is 12240cm.

ii. (c) 27

Solution:

Here $594 = 2 \times 33 \times 11$ and $189 = 33 \times 7$

HCF of 594 and 189 = $3^3 = 27$

Hence, the maximum number of columns in which they can march is 27.

iii. (c) 12 litres

Solution:

Here $768 = 28 \times 3$ and $420 = 22 \times 3 \times 5 \times 7$

HCF of 768 and 420 = $22 \times 3 = 12$

So, the container which can measure fuel of either tanker exactly must be of 12 litres.

iv. (b) 75cm

Solution:

Here, Length = 825cm, Breadth = 675cm and Height = 450cm

Also, $825 = 5 \times 5 \times 3 \times 11$, $675 = 5 \times 5 \times 3 \times 3 \times 3$ and $450 = 2 \times 3 \times 3 \times 5 \times 5$

HCF = $5 \times 5 \times 3 = 75$

Therefore, the length of the longest rod which can measure the three dimensions of the room exactly is 75cm.

v. (a) 3 and 2

Solution:

LCM of 8 and 12 is 24.

$\therefore$ The least number of pack of pens = $\frac{24}{8} = 3$

$\therefore$ The least number of pack of note pads = $\frac{24}{12} = 2$

Assertion Reason Answer-

1. (a) Both A and R are true and R is the correct explanation of A.
2. (c) A is true but R is false.

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